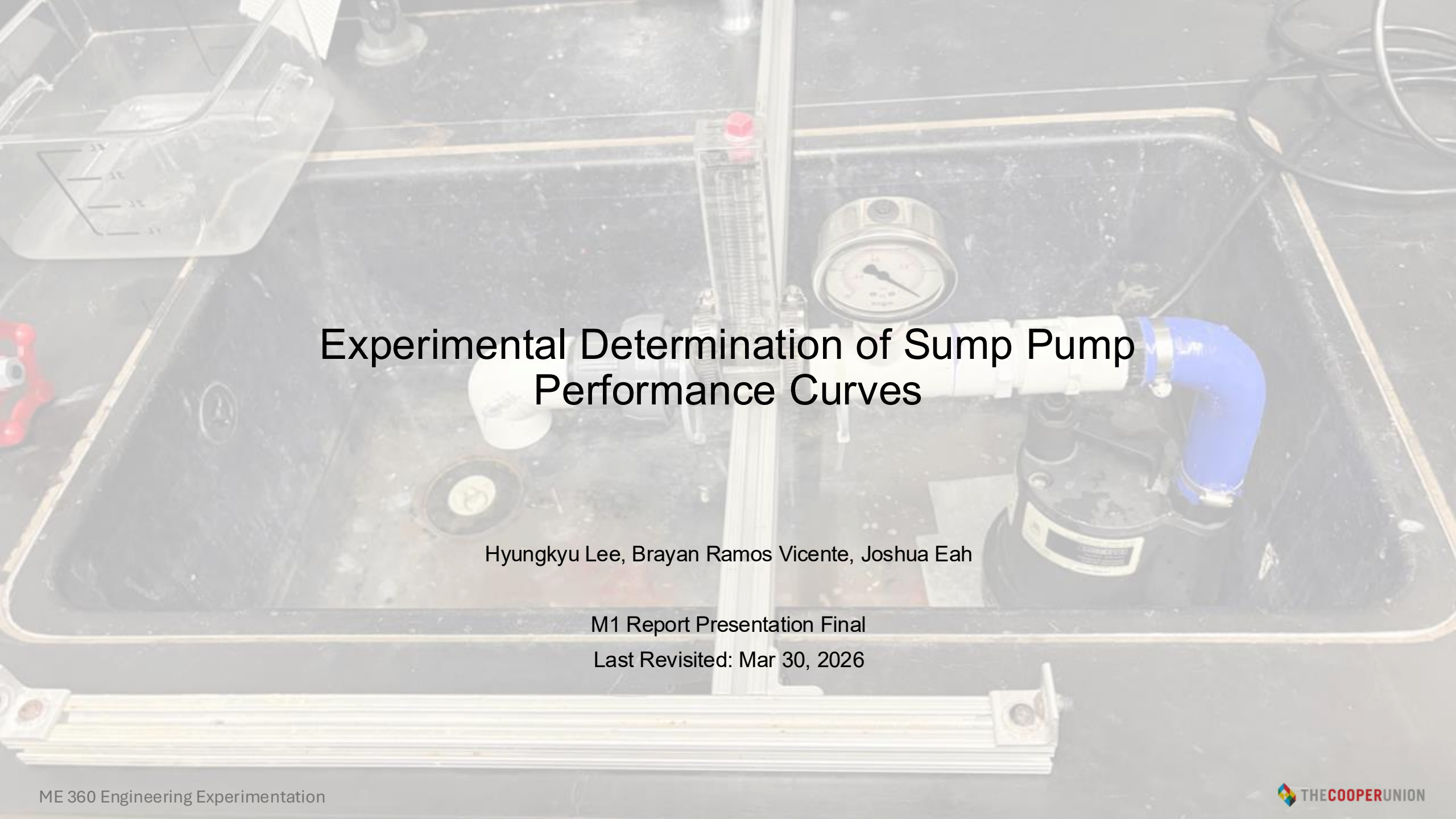


Experimental Determination of Sump Pump Performance Curves

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M1 Report Presentation Final

Last Revisited: Mar 30, 2026

Floods often occur due to lack of drainage or natural disasters. This excess of water reaches sump pits such as basements and crawlspaces.



Figure: Flooded Crawlspace in Foundation of Home[1)]

1) https://www.afsrepair.com/resources/crawl-space-repair/how-to-deal-with-a-flooded-crawl-space/

Sump pumps are centrifugal pumps with uses in sump basins to avoid flooding.

- Sump pumps collect excess water from its surroundings and pump it away.
- Works automatically once a certain level of water is detected, making it convenient and efficient.



Figure: Sump Pump Placement[1)]

2) https://www.onpointnyct.com/2025/04/03/everything-you-need-to-know-about-sump-pumps-and-basement-drainage-systems/

Centrifugal pumps use a rotating impeller (motor) that increases speed and pressure of a liquid, causing it to be pumped.

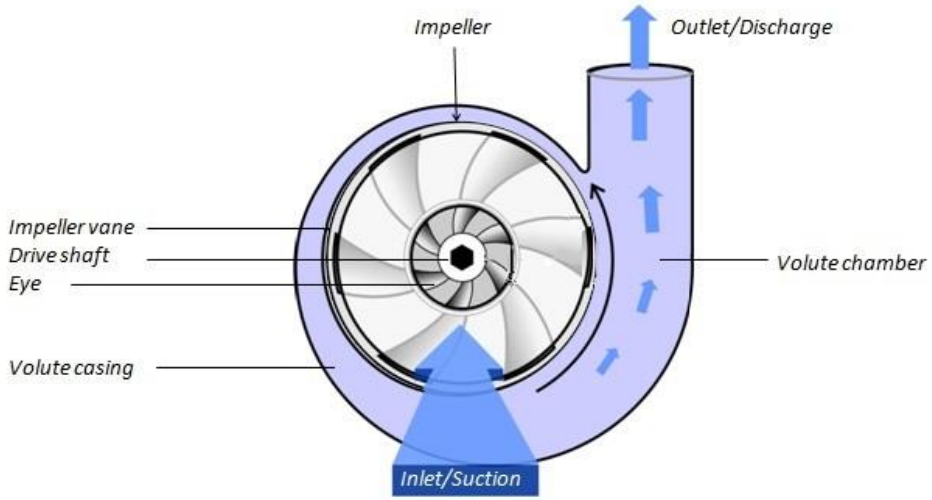


Figure: Centrifugal Pump Diagram[1)]

- Fluid enters through the eye of the impeller
- Due to rotations, centrifugal force accelerates fluid through impeller vanes



Figure: Everbilt 1/6 HP Plastic Utility Pump[2)]

- Nominal operating voltage: 115V
- Max head: 20 ft
- Compatible media: Water

1) https://www.michael-smith-engineers.co.uk/resources/useful-info/centrifugal-pumps
2) https://images.thdstatic.com/productImages/590b8033-153a-4a1d-8d82-110bb826a74b/svn/everbilt-submersible-utility-pumps-sup54-hd-64_1000.jpg

This experiment measures how a sump pump responds to changing system loads.

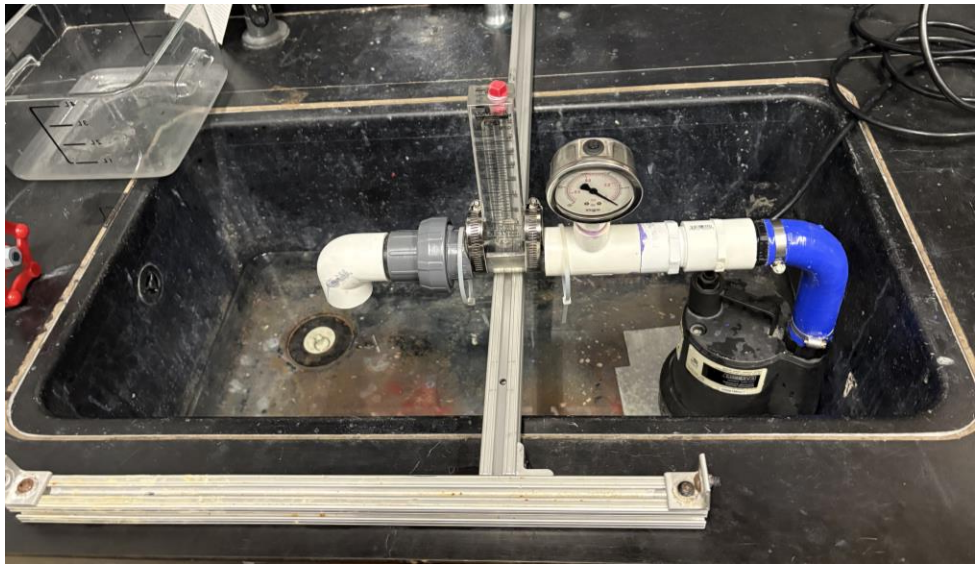


Figure: Picture of Experimental Setup

Primary output: Pump performance map relating

- pressure head vs flow rate
- power vs flow rate

Pump load is varied using a regulating valve in a closed-loop test stand.

Data collected at a fixed input AC voltage.

The pump test stand recirculates water through a closed loop in which system resistance can be varied using a regulating valve.

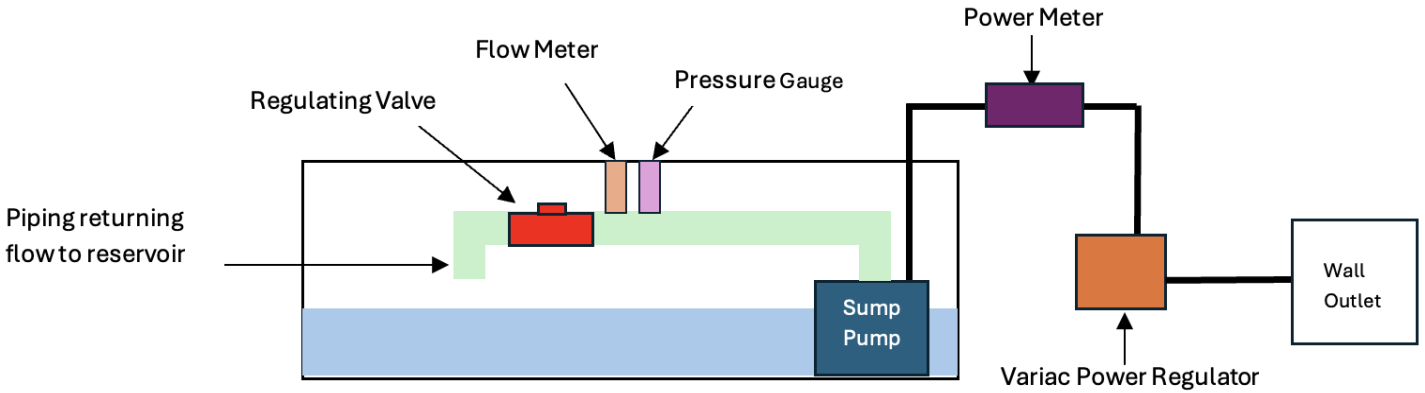


Figure: Diagram of Apparatus

Measurement points taken with

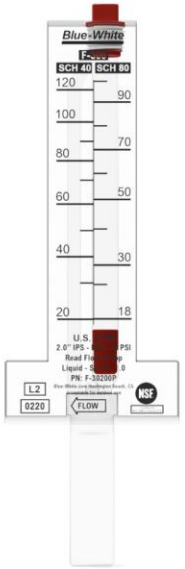
1. Flow Meter
2. Pressure Gauge
3. Power Meter

Sump pump is connected to Variac Power Regulator (power meter), which is connected to an AC wall outlet.

Measurements were taken with a F-300 Pilot Tube Flow Meter, NoShok Pressure Gauge, and a “Watts Up” Power Meter.

F-300 Pilot Tube Flow Meter

- Max. Fluid Temp: 190°F / 88°C at 0 psi
- Max. working pressure: 75 psi at 70°F / 21°C



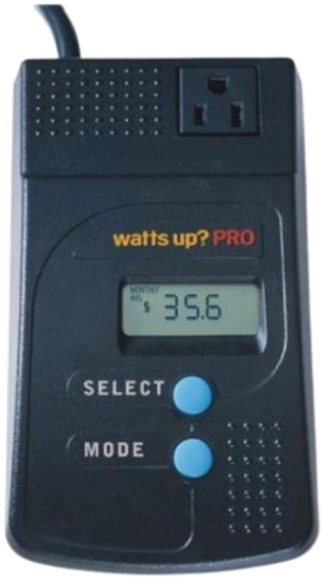
NoShok 0-15 psi Pressure Gauge

- Pressure Range: 0-15 psi
- Ambient Temp.: -4-140°F
- Accuracy: 1.6%



“Watts Up” Power Meter

- Voltage Range: 0-120V
- Maximum Current: 15A
- Frequency: 60Hz
- Accuracy: ±3% for loads above 10W



1) https://www.blue-white.com/product/f-300-pitot-tube-acrylic-flow-meter/
2) https://www.zoro.com/noshok-pressure-gauge-0-to-15-psi-psi-2-12-in-dial-14-in-npt-or-716-in-20-adjustable-npt-25-901-15-psikpa/i/G608048869/
3) https://www.powermeterstore.com/p1206/watts_up_pro.php

Pump performance data was collected by adjusting the regulating valve and recording steady-state operating conditions.

1. Values with valve fully closed and without valve
2. Adjust the valve to obtain measurements of pressure, flow rate, and power consumption



Figure: Regulating Valve on Pump Rig

Every team was assigned a unique Variac Setting (voltage) for this experiment.

Team	Variac Setting
A1	125 V
A2	110 V
A3	95 V
B4	100 V
B5	115 V
B6	120 V (i)
A7	120 V (ii)

- Data points were recorded for each Variac setting.
- Various voltages were tested to find the optimal point for efficiency based on voltage in efforts to create pump performance maps.

Pump performance maps showcase various head, efficiencies, and power depending on flow rate. This displays the Best Efficiency Point of a pump.

- Pump performance maps are used for troubleshooting and operating pumps
- The best efficiency point (BEP) is the point at which the pump's efficiency is highest and where the head conditions and flow rate are optimal.

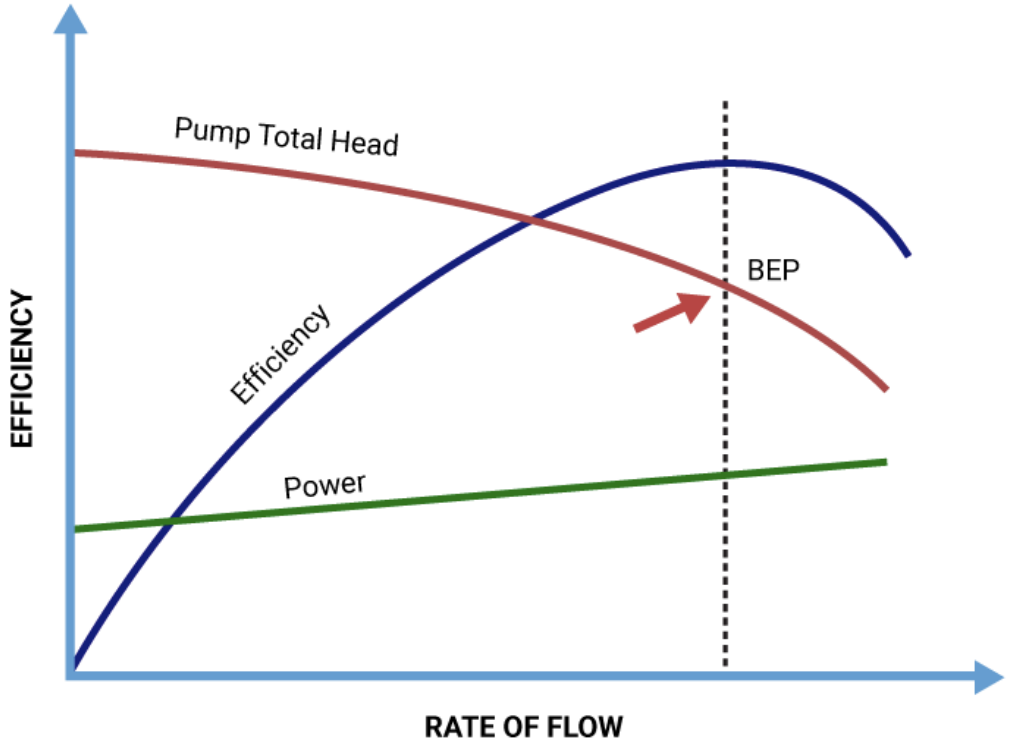
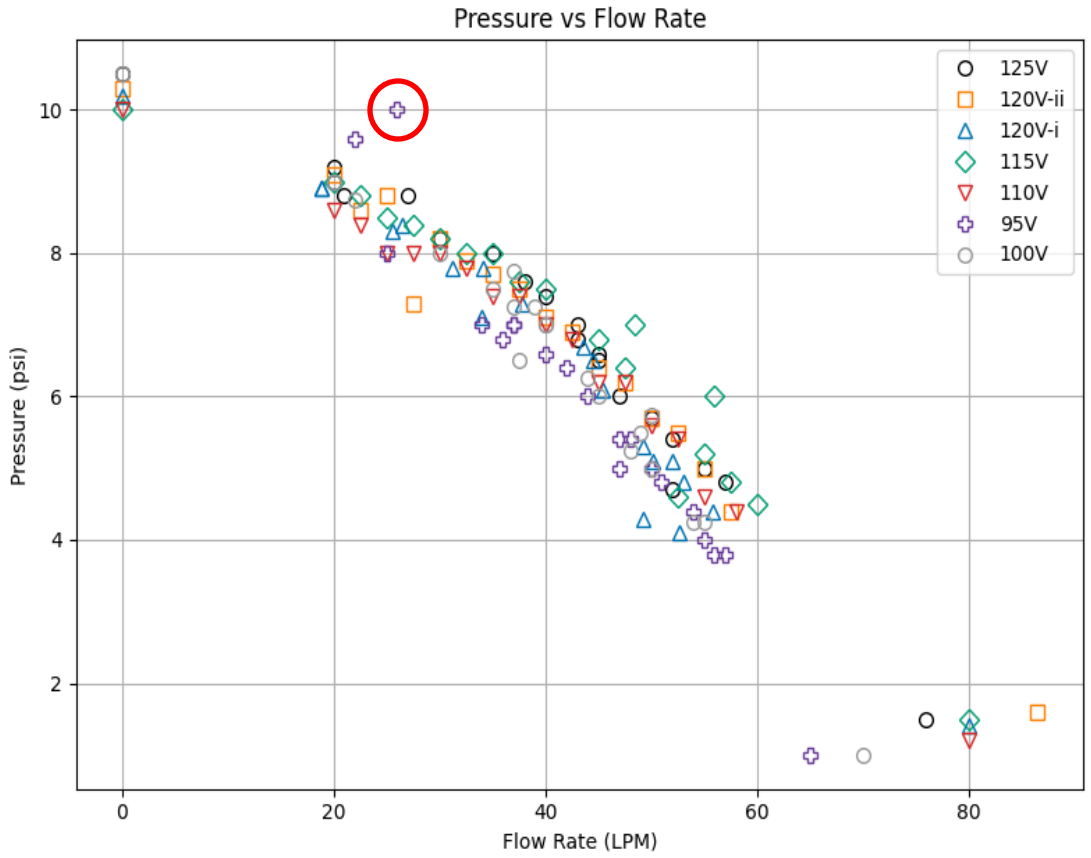
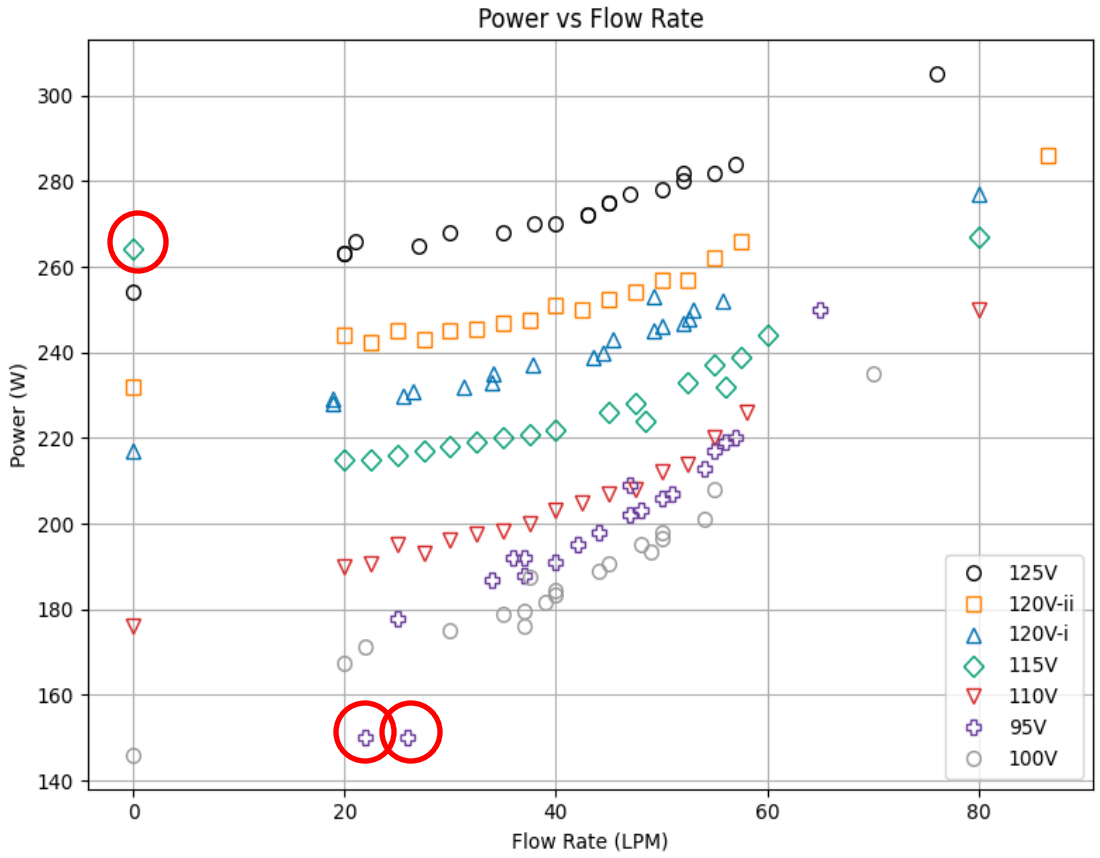


Figure: Pump performance map of a pump[1)]

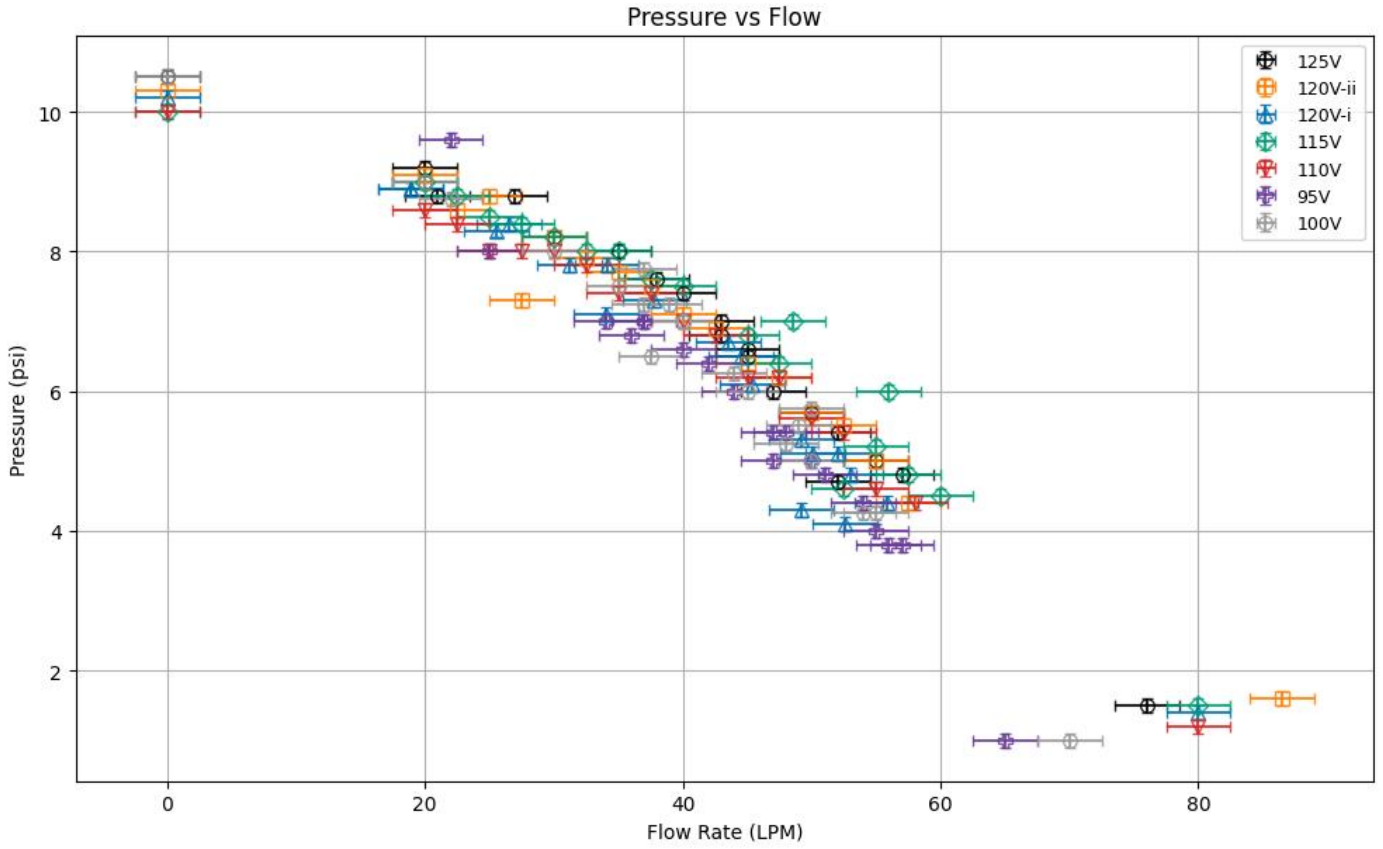
1) https://home.pumpsystemsacademy.com/blog/understanding-pump-curves-guide-for-engineers

Four data points were removed as outliers prior to creating the model.



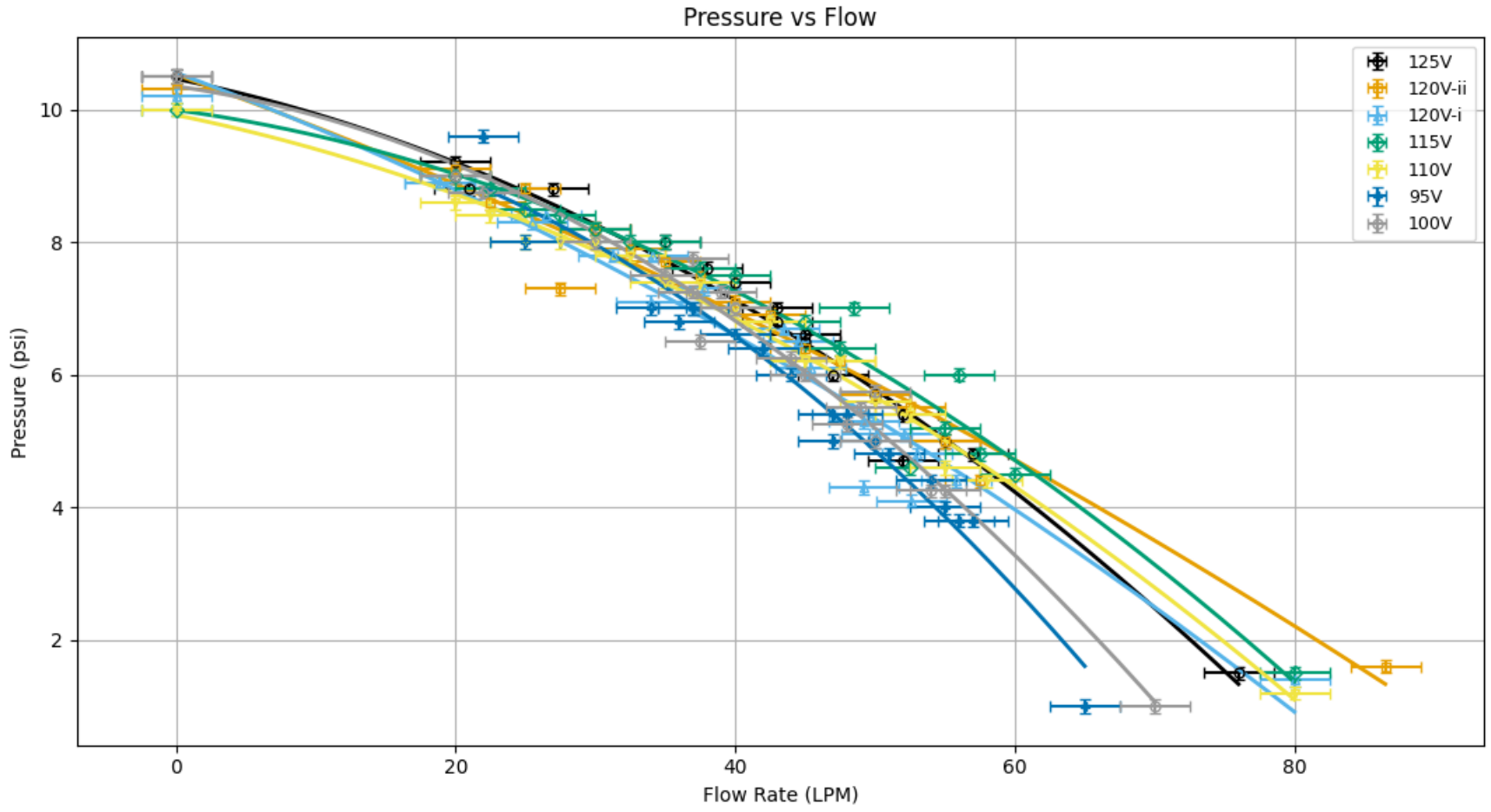


As the flow rate increases, the pressure decreases.



- All trials show very similar results
- Results should be dependent only on flow geometry and fluid (fluid dynamics)
- Only the flow characteristics of the piping in the rig are reflected

Quadratic polynomial models accurately capture the flow rate–pressure behavior across all Variac settings, with high R^2 values indicating strong agreement.



Pressure vs Flow (Quadratic Fit)

125V:
$R^2 = 0.9850$
$P(Q) = (-0.0010173)Q^2 + (-0.042782)Q + (10.462)$

120V-ii:
$R^2 = 0.9719$
$P(Q) = (-0.00035668)Q^2 + (-0.075401)Q + (10.526)$

120V-i:
$R^2 = 0.9567$
$P(Q) = (-0.00053502)Q^2 + (-0.077676)Q + (10.554)$

115V:
$R^2 = 0.9641$
$P(Q) = (-0.00099095)Q^2 + (-0.028561)Q + (9.9918)$

110V:
$R^2 = 0.9909$
$P(Q) = (-0.00084048)Q^2 + (-0.042954)Q + (9.9164)$

95V:
$R^2 = 0.9663$
$P(Q) = (-0.0017534)Q^2 + (-0.014973)Q + (9.988)$

100V:
$R^2 = 0.9813$
$P(Q) = (-0.0014722)Q^2 + (-0.029503)Q + (10.349)$

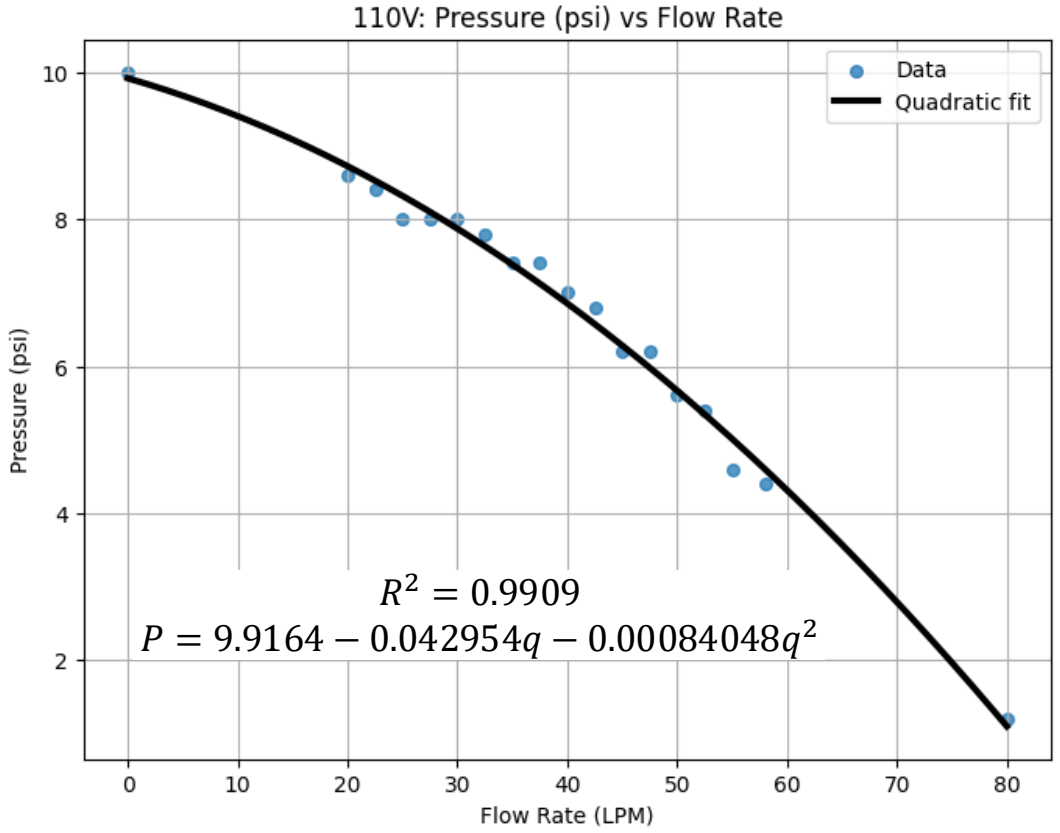
Head vs. Flow model can be derived from fluid mechanics.

- Euler's Pump Equation:
  - $H_{pump} = \frac{U_{out}V_{out} - U_{in}V_{in}}{g}$
- The fluid enters the pump straight, $V_{in} = 0$
  - $H_{pump} = \frac{U_{out}V_{out}}{g}$
- Getting flow rate:
  - $V_{out} = U_2 - V_{r_{out}} \cot(\beta_{out})$
  - $V_{r_{out}} = \frac{Q}{\pi D b}$

Head vs. Flow model can be derived from fluid mechanics.

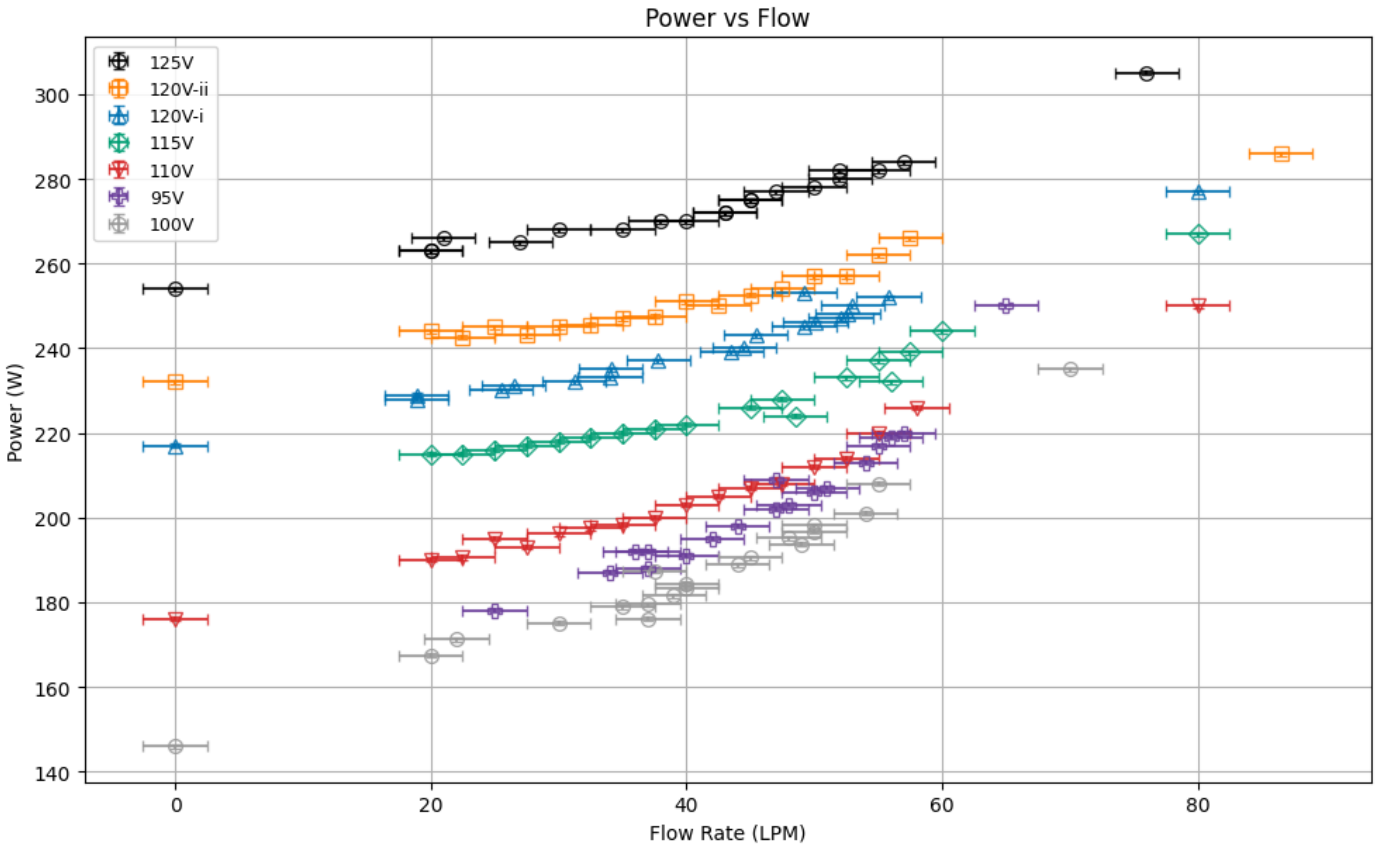
- Head-flow relationship:
  - $H_{pump} = \frac{U_{out}^2}{g} - \left[\frac{U_{out}\cot(\beta_{out})}{g\pi Db}\right] Q$
- Frictional losses:
  - $h_f = f\frac{L}{D}\frac{v^2}{2g}$
- Thus, final generalized form:
  - $\boldsymbol{H = A - BQ - CQ^2}$

Pressure curves follow expectations from the fluid dynamics model.



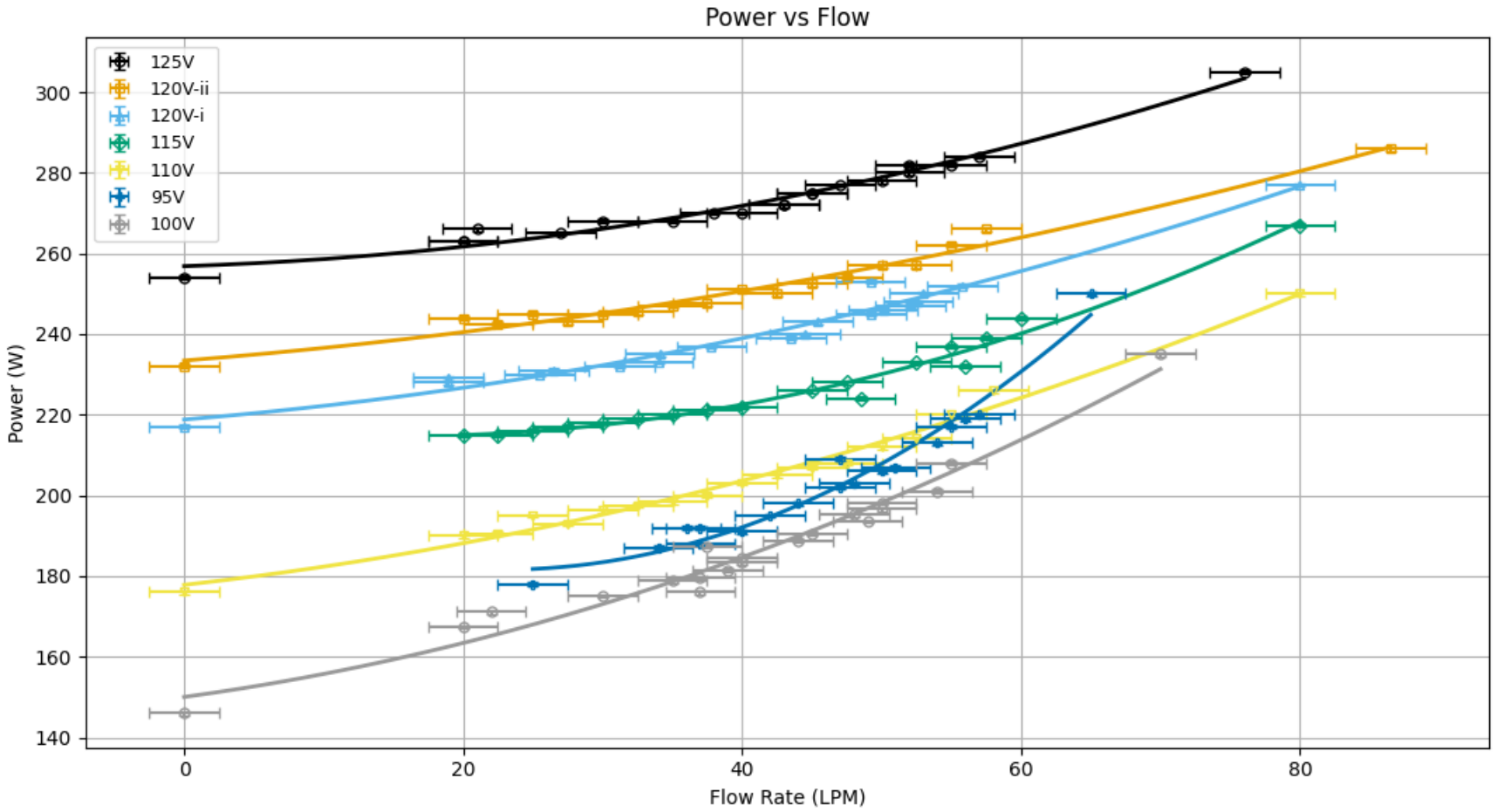
- Quadratic model has excellent match to data over entire range
- Constants represent physical parameters:
  - A: shut-off pressure
  - B: combined linear effects of pressure vs flow in Euler model and minor (linear) losses
  - C: dynamic losses in fluid system

As the flow rate increases, the power increases.



- Theory $P = \rho g Q H$
- The relationship is can be approximated by a linear trend, though a quadratic relationship can give better results.
- Power usage can differ significantly for a given flow rate

Quadric polynomial models accurately capture the flow rate–power behavior across all Variac settings, with high R^2 values indicating strong agreement with experimental data.



Power vs Flow (Quadratic Fit)

125V:
$R^2 = 0.9772$
$W(Q) = (0.0066135)Q^2 + (0.10928)Q + (256.85)$

120V-ii:
$R^2 = 0.9767$
$W(Q) = (0.0039078)Q^2 + (0.27431)Q + (233.43)$

120V-i:
$R^2 = 0.9721$
$W(Q) = (0.0054857)Q^2 + (0.28423)Q + (218.81)$

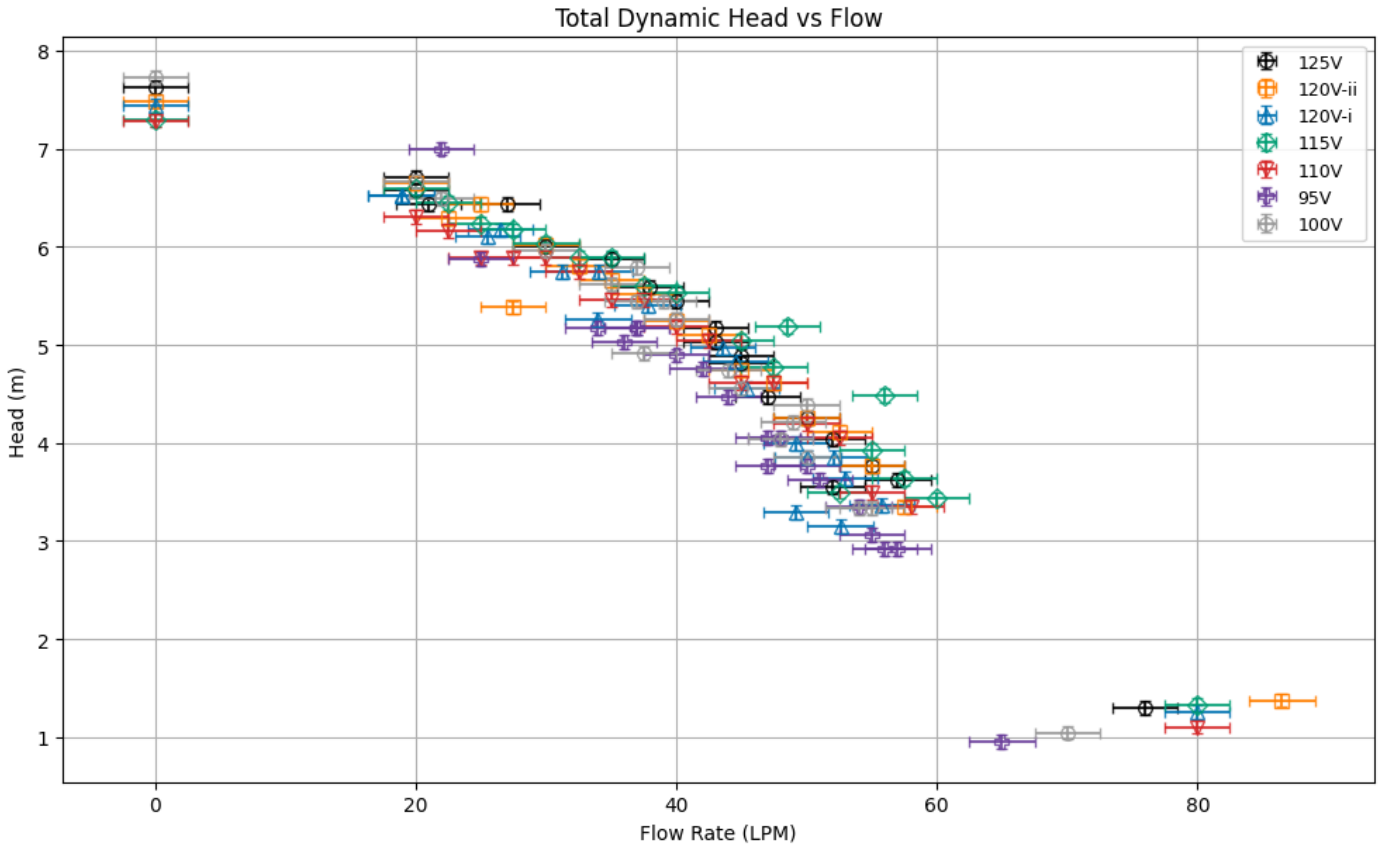
115V:
$R^2 = 0.9791$
$W(Q) = (0.012625)Q^2 + (-0.38075)Q + (217.53)$

110V:
$R^2 = 0.9874$
$W(Q) = (0.0064328)Q^2 + (0.38721)Q + (177.82)$

95V:
$R^2 = 0.9672$
$W(Q) = (0.035482)Q^2 + (-1.6178)Q + (200.03)$

100V:
$R^2 = 0.9698$
$W(Q) = (0.0098518)Q^2 + (0.4718)Q + (150.02)$

Total Dynamic Head (TDH) is the total energy (per unit weight of fluid) that a pump must provide to move water through a system.



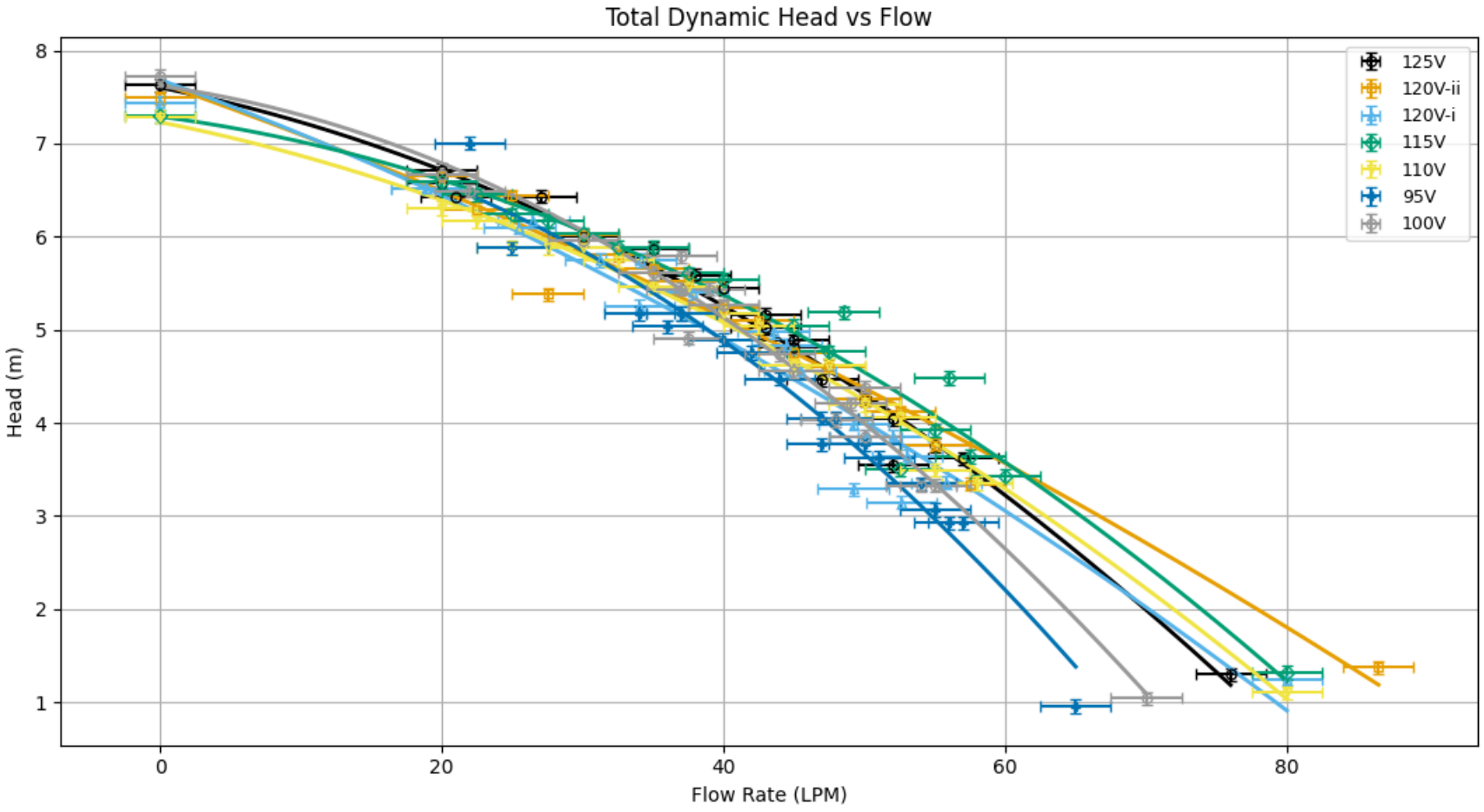
Calculating Total Dynamic Head

$$H_p = \frac{\Delta p}{\rho g} + z_{cl}$$

Sample Calculation:

$$H_p = \frac{48263.32\text{Pa}}{\left(\frac{1000kg}{m^3}\right)\left(\frac{9.81m}{s^2}\right)} + 0.26m = 5.18m$$

Across all voltage trials TDH decreases quadratically with increasing flow rate.



Total Dynamic Head vs Flow (Quadratic Fit)

125V:
$R^2 = 0.9850$
$H(Q) = (-0.00071499)Q^2 + (-0.030068)Q + (7.601)$

120V-ii:
$R^2 = 0.9719$
$H(Q) = (-0.00025068)Q^2 + (-0.052994)Q + (7.6488)$

120V-i:
$R^2 = 0.9567$
$H(Q) = (-0.00037603)Q^2 + (-0.054593)Q + (7.6847)$

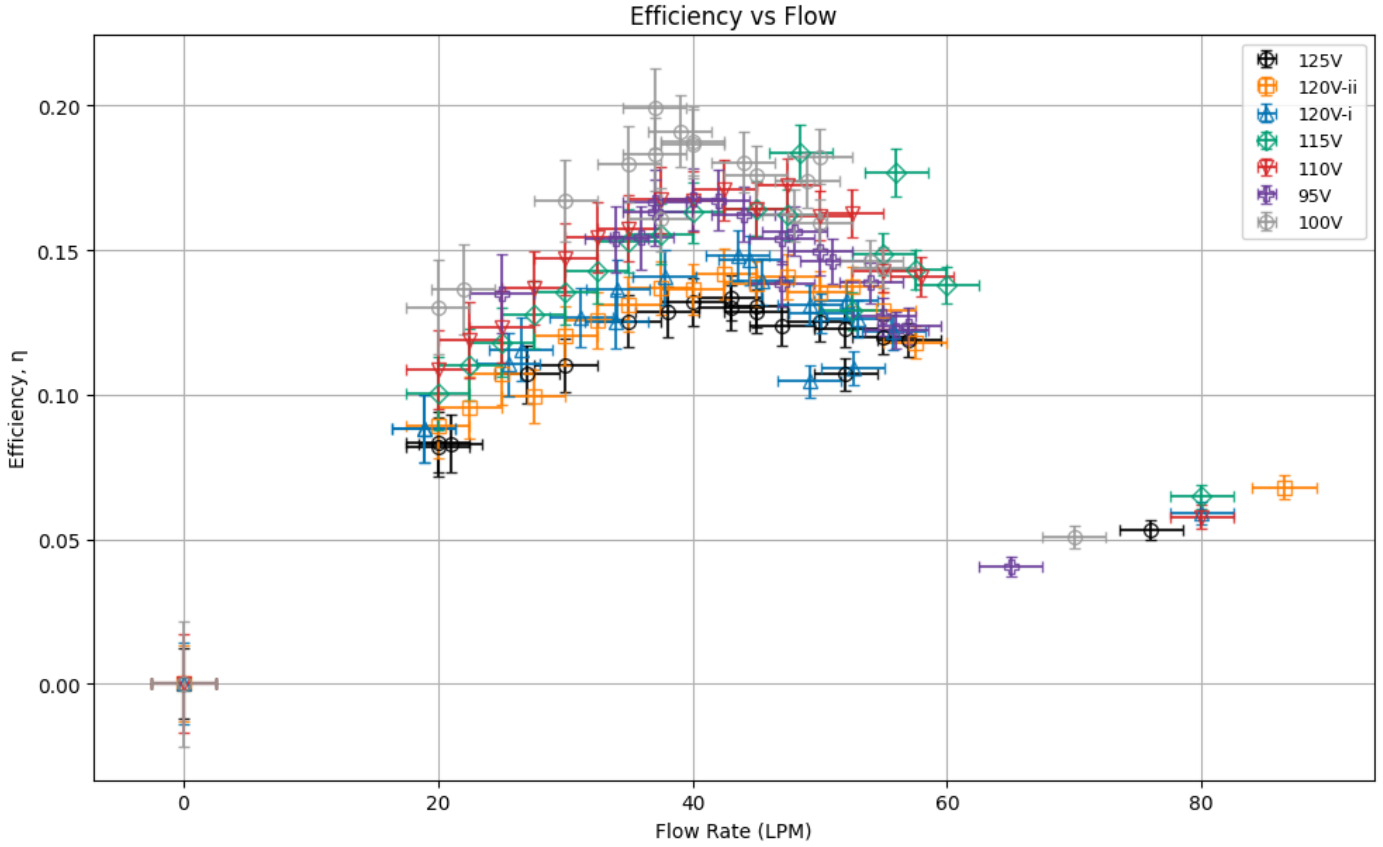
115V:
$R^2 = 0.9641$
$H(Q) = (-0.00069647)Q^2 + (-0.020074)Q + (7.2892)$

110V:
$R^2 = 0.9909$
$H(Q) = (-0.00059071)Q^2 + (-0.030189)Q + (7.2299)$

95V:
$R^2 = 0.9663$
$H(Q) = (-0.0012324)Q^2 + (-0.010524)Q + (7.2739)$

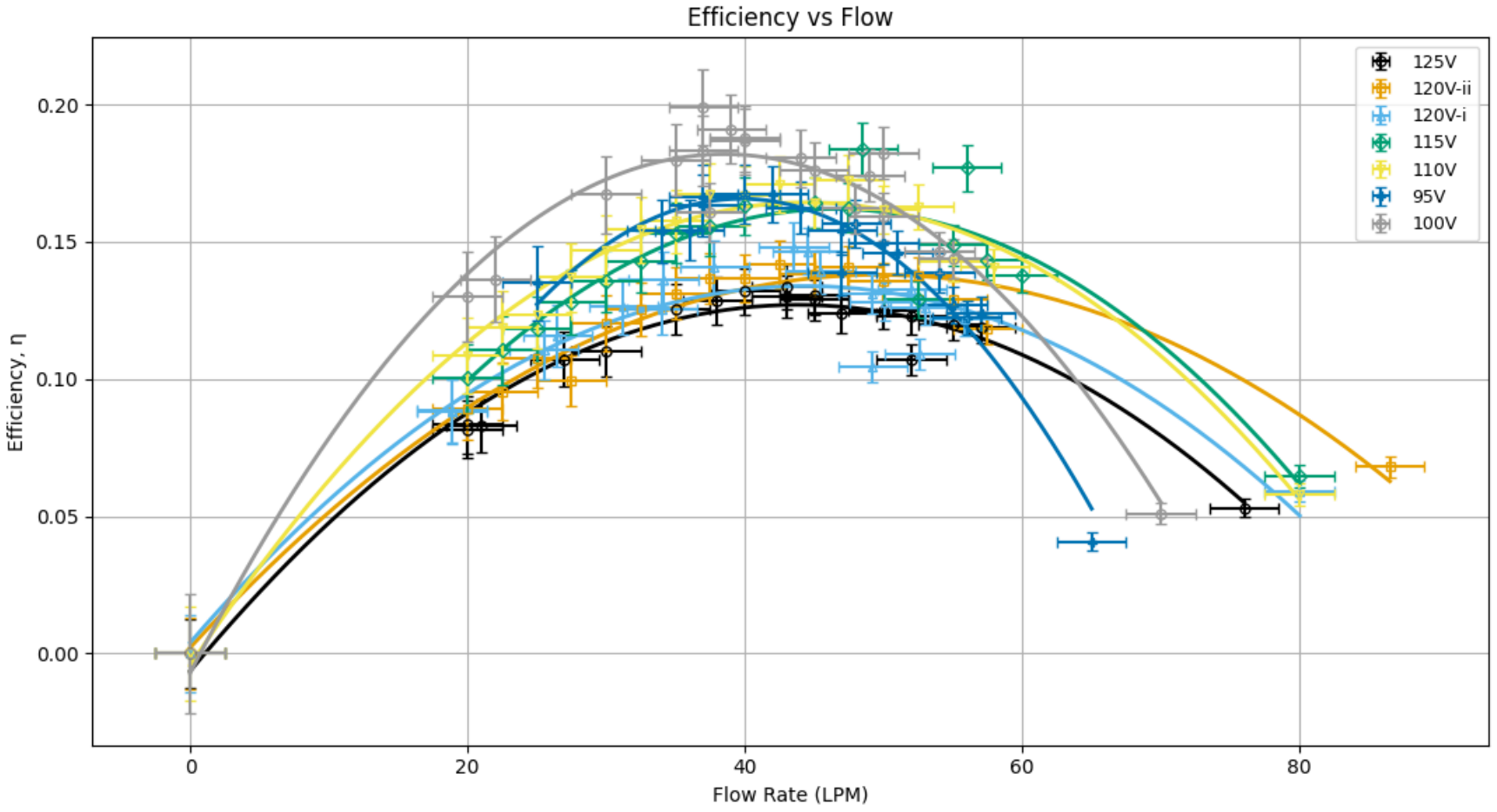
100V:
$R^2 = 0.9813$
$H(Q) = (-0.0010347)Q^2 + (-0.020736)Q + (7.6168)$

Pump efficiency measures how effectively electrical power is converted into water movement.



- Efficiency is computed by comparing hydraulic power at a given electrical input power level
- Sample Calculation
  - $\eta = \frac{\mathrm{P_{hyd}}}{\mathrm{P_{elec}}} = \frac{33.9\mathrm{W}}{203\mathrm{W}} = 16.7\%$

Across all voltage trials, efficiency follows a quadratic trend with flow rate, peaking at mid-range flows and decreasing at both low and high operating conditions.



Efficiency vs Flow (Quadratic Fit)

125V:
$R^2 = 0.9731$
$\eta(Q) = (-6.9525e\text{-}05)Q^2 + (0.006091)Q + (-0.006362)$

120V-ii:
$R^2 = 0.9684$
$\eta(Q) = (-5.5217e\text{-}05)Q^2 + (0.0054766)Q + (0.0021804)$

120V-i:
$R^2 = 0.9152$
$\eta(Q) = (-6.5944e\text{-}05)Q^2 + (0.0058541)Q + (0.0039636)$

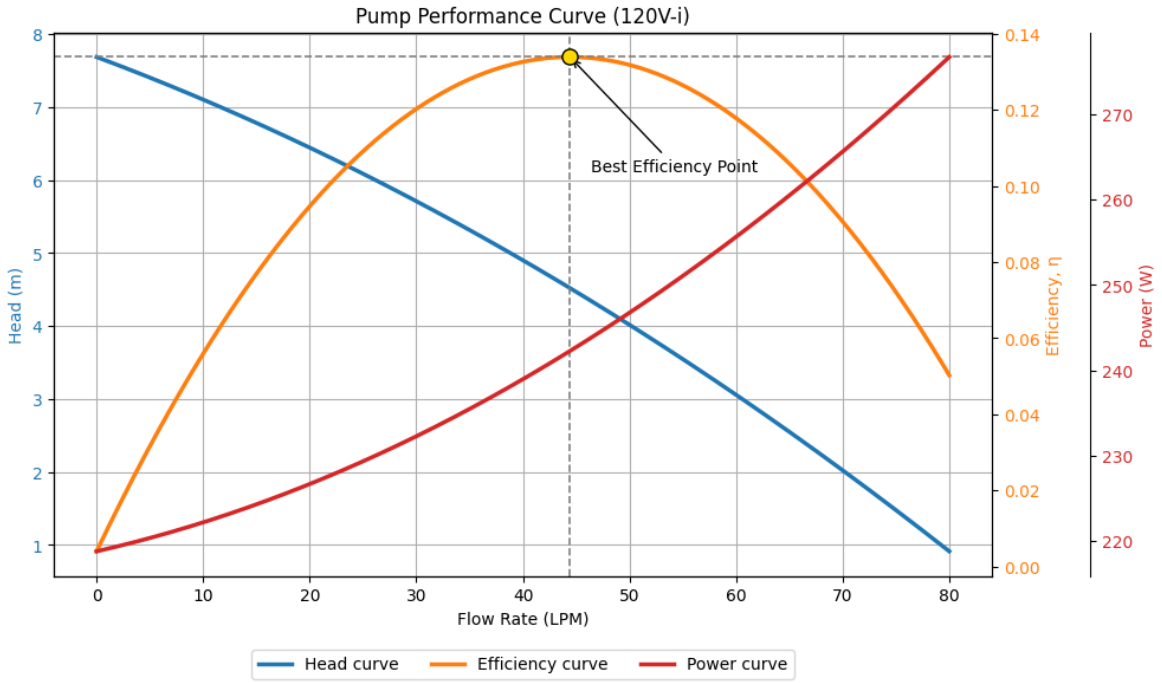
115V:
$R^2 = 0.8503$
$\eta(Q) = (-8.9259e\text{-}05)Q^2 + (0.008296)Q + (-0.03081)$

110V:
$R^2 = 0.9830$
$\eta(Q) = (-8.5472e\text{-}05)Q^2 + (0.0075931)Q + (-0.004308)$

95V:
$R^2 = 0.9388$
$\eta(Q) = (-0.00017695)Q^2 + (0.014064)Q + (-0.11379)$

100V:
$R^2 = 0.9632$
$\eta(Q) = (-0.0001277)Q^2 + (0.0098262)Q + (-0.0071062)$

Pump Performance Curves

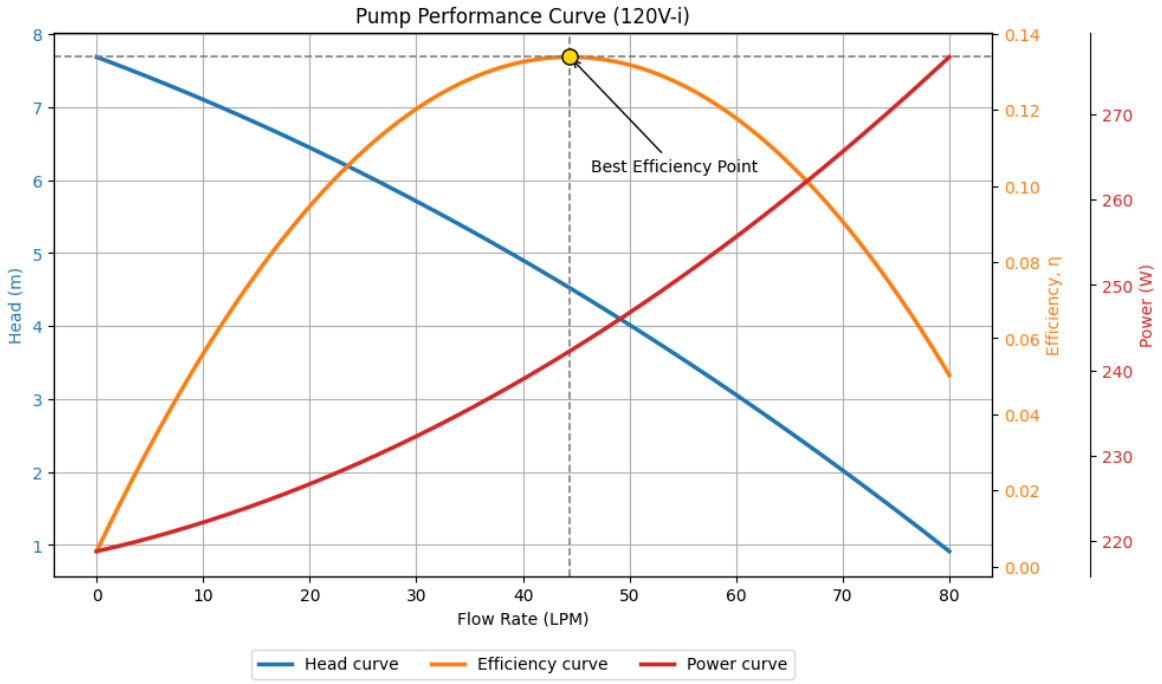


HEAD VS FLOW
LINE OF BEST FIT EQUATION:
H(q) = (-0.00037603)q^2 + (-0.0545927)q + (7.68466)
UNCERTAINTY OF EACH COEFFICIENT:
a uncertainty = ±0.000161974
b uncertainty = ±0.0133122
c uncertainty = ±0.274015

EFFICIENCY VS FLOW
LINE OF BEST FIT EQUATION:
η(q) = (-6.59443e-05)q^2 + (0.00585406)q + (0.0039636)
UNCERTAINTY OF EACH COEFFICIENT:
a uncertainty = ±5.26813e-06
b uncertainty = ±0.000432973
c uncertainty = ±0.0089122

POWER VS FLOW
LINE OF BEST FIT EQUATION:
W(q) = (0.00548566)q^2 + (0.284225)q + (218.808)
UNCERTAINTY OF EACH COEFFICIENT:
a uncertainty = ±0.00110753
b uncertainty = ±0.0910251
c uncertainty = ±1.87363

Computing the BEP (best efficiency point)



- **BEP is computed by finding the apex of the curve using the 1st derivative**
- Take regression as "true fit", using equation:

$$\eta = c + bq + aq^2$$

- Derivative yields $\mathrm{q} = -\frac{b}{2a}$
- In the 120V-I case:
  - $q = -\frac{0.005854}{2*-0.00006594} = 44.39\ LPM$
  - BEP: $(44.39, y)$
    - $y = 0.1339 = \mathbf{13.39\%}$
  - Uncertainty (a, b, c are dependent):

$$\sigma_{bep}^2$$

$$= \left(\frac{d\eta}{da}\right)\sigma_a^2 + \left(\frac{d\eta}{db}\right)\sigma_b^2 + \left(\frac{d\eta}{dc}\right)\sigma_c^2 + 2\left(\frac{d\eta}{da}\frac{d\eta}{db}\right)Cov(a,b) + 2\left(\frac{d\eta}{da}\frac{d\eta}{dc}\right)Cov(a,c) + 2\left(\frac{d\eta}{db}\frac{d\eta}{dc}\right)Cov(b,c)$$

Efficiency tends to peak at pump design voltage.

- BEP determined for each test case
- Highest efficiency achieved within nominal AC voltage range (i.e. ~110V)
- Exceeding this range leads to drops in efficiency and flow rate
- However, flow rate @ BEP and efficiency not balanced

Case	BEP
120V (i)	44.39 ± 1.1621LPM, 13.39 ± 0.002808%
110V	44.42 ± 0.5199 LPM, 16.43 ± 0.001746%
95V	39.72 ± 0.7185 LPM, 16.54 ± 0.002385%
115V	46.47 ± 1.0300 LPM, 16.2 ± 0.003756%
120 (ii)	49.59 ± 0.8601LPM, 13.8 ± 0.001993%
100V	38.47 ± 0.6246 LPM, 18.19 ± 0.002635%
125V	43.81 ± 0.6391LPM, 12.71 ± 0.001519%

Computing specific energy consumption highlights actual energy use.

- $SEC = \frac{P_{in}}{Q}$
  - i.e. power per unit flow
- Highlights efficiency of pumping (accounts for flow rate & power use)

Voltage (V)	Flow at BEP (LPM)	Power at BEP (W)	SEC (W/LPM) (lower is better)	Head (m)
100V	38.47 ± 0.62	182.8 ± 7.4	4.75 ± 0.21	5.288 ± 0.471
110V	44.42 ± 0.52	207.7 ± 4.5	4.68 ± 0.11	4.723 ± 0.328
115V	46.47 ± 1.03	227.1 ± 8.8	4.89 ± 0.22	4.852 ± 0.700
120V (i)	44.39 ± 1.16	242.2 ± 5.0	5.46 ± 0.18	4.520 ± 0.733

Computing specific energy consumption highlights actual energy use.

- 110V case yields best nominal SEC
  - Not statistically distinct from 100V case
- 100V gives good SEC and has higher nominal head at BEP
- Overall data is statistically inconclusive; nominal values suggest general trends

Voltage (V)	Flow at BEP (LPM)	Power at BEP (W)	SEC (W/LPM) (lower is better)	Head (m)
100V	38.47 ± 0.62	182.8 ± 7.4	4.75 ± 0.21	5.288 ± 0.471
110V	44.42 ± 0.52	207.7 ± 4.5	4.68 ± 0.11	4.723 ± 0.328
115V	46.47 ± 1.03	227.1 ± 8.8	4.89 ± 0.22	4.852 ± 0.700
120V (i)	44.39 ± 1.16	242.2 ± 5.0	5.46 ± 0.18	4.520 ± 0.733

Pump performance trends follow theoretical expectations across flow rate and voltage.

1. Pressure (Head) Decreases as Flow Increases
   Inline with theory $H = H_0 - kQ^2$
2. Power Increases as Flow Increases
   Inline with theory $P = \rho g Q H$
3. Higher Voltage Shifts Curves Upward
   Increasing voltage increases maximum head and flow
4. Pump Performance Follows Expected Quadratic Behavior
   Head vs. flow is well modeled by a quadratic relationship with high R^2 values
5. Best Efficiency Occurs at Mid-Range Flow Rates
   BEP found to be at 38–47 LPM, however efficiency is generally poor in all domains

Observed deviations and uncertainties are primarily driven by operating conditions and measurement limitations.

1. The 95V Trial is Questionable
   Motor may be near stall region
2. 115V trial not as accurate compared to manufacturer data (60 LPM vs 76.47 LPM[1)] for ~3m head)
   Difference in testing conditions (setup, wear on pump, sensor accuracy)

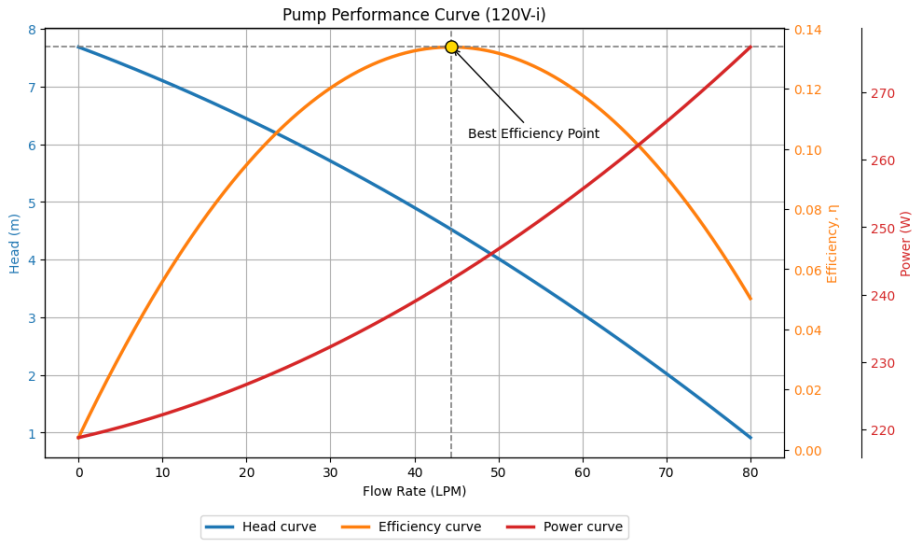
1) https://images.thdstatic.com/catalog/pdfImages/b5/b55aeecf-36e8-4149-8e39-48c07681db51.pdf

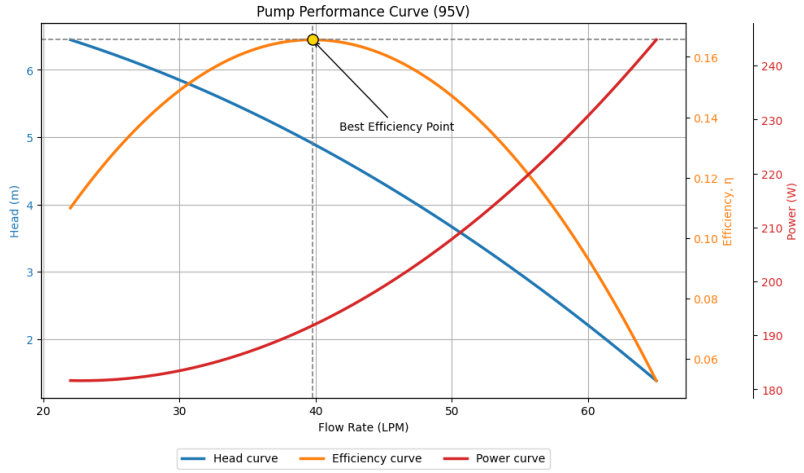
Measurement error and system instability were the primary sources of uncertainty; addressing these through improved instrumentation and control would enhance result accuracy.

1. Better Instrumentation
   - Higher resolution flow meter to reduce ±5 LPM uncertainty
   - Digital pressure sensor
   - Real time recording of data can normalize for fluctuations
2. Better Method of Control
   - Replace manual valve adjustments with fine control valve
   - Allow system to reach steady state before recording data

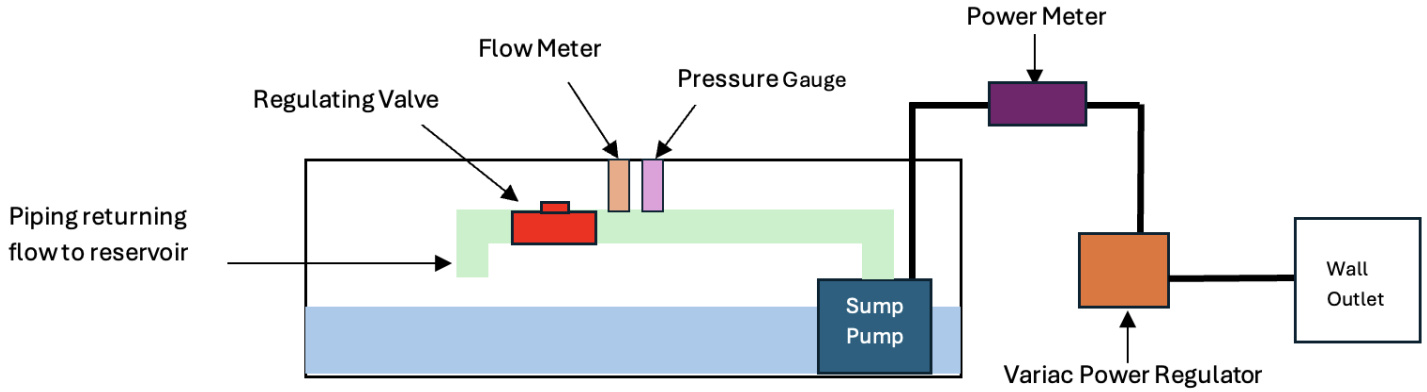
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Pump Performance Curve (120V-i)
Best Efficiency Point
Head (m)
Efficiency, η
Power (W)
Flow Rate (LPM)
Head curve
Efficiency curve
Power curve


Pump Performance Curve (95V)
Best Efficiency Point
Head (m)
Efficiency, η
Power (W)
Flow Rate (LPM)
Head curve
Efficiency curve
Power curve

Thaηk yoμ


Power Meter
Flow Meter
Pressure Gauge
Regulating Valve
Piping returning
flow to reservoir
Sump
Pump
Wall
Outlet
Variac Power Regulator

[Backup] Raw Data

120 V - i			
#	Flow Rate (LPM)	Pressure (psi)	Power (Watts)
1	0	10.2	217
2	18.9	8.9	228
3	18.9	8.9	229
4	25.5	8.3	230
5	26.5	8.4	231
6	31.23	7.8	232
7	34	7.1	233
8	34.1	7.8	235
9	37.8	7.3	237
10	43.53	6.7	239
11	44.5	6.5	240
12	45.42	6.1	243
13	49.21	5.3	245
14	50.1	5.1	246
15	52.05	5.1	247
16	52.6	4.1	248
17	53	4.8	250
18	55.8	4.4	252
19	49.2	4.3	253
20	80	1.4	277

120 V - ii			
#	Flow Rate (LPM)	Pressure (psi)	Power (Watts)
1	86.5	1.6	286
2	0	10.3	232
3	20	9.1	244
4	25	8.8	245
5	30	8.2	245
6	35	7.7	247
7	40	7.1	251
8	45	6.4	252.5
9	50	5.7	257
10	55	5.0	262
11	57.5	4.4	266
12	52.5	5.5	257
13	47.5	6.2	254
14	42.5	6.9	250
15	37.5	7.5	247.5
16	32.5	7.9	245.5
17	27.5	7.3	243
18	22.5	8.6	242.5

115 V			
#	Flow Rate (LPM)	Pressure (psi)	Power (Watts)
1	80	1.5	267
2	57.5	4.8	239
3	52.5	4.6	233
4	60	4.5	244
5	55	5.2	237
6	56	6	232
7	45	6.8	226
8	40	7.5	222
9	35	8	220
10	30	8.2	218
11	25	8.5	216
12	20	9	215
13	22.5	8.8	215
14	27.5	8.4	217
15	32.5	8	219
16	37.5	7.6	221
17	48.5	7	224
18	47.5	6.4	228
19	0	10	264

110 V			
#	Flow Rate (LPM)	Pressure (psi)	Power (Watts)
1	80	1.2	250
2	0	10	176
3	30	8	196.2
4	40	7	203
5	50	5.6	212
6	55	4.6	220
7	58	4.4	226
8	45	6.2	207
9	35	7.4	198.4
10	25	8	195
11	20	8.6	190
12	22.5	8.4	190.6
13	27.5	8	193
14	32.5	7.8	197.5
15	37.5	7.4	200
16	42.5	6.8	205
17	47.5	6.2	208
18	52.5	5.4	214

95 V			
#	Flow Rate (LPM)	Pressure (psi)	Power (Watts)
1	65	1	250
2	26	10	150
3	47	5	209
4	37	7	192
5	22	9.6	150
6	36	6.8	192
7	42	6.4	195
8	54	4.4	213
9	56	3.8	219
10	57	3.8	220
11	51	4.8	207
12	40	6.6	191
13	50	5.0	206
14	55	4.0	217
15	25	8.0	178
16	34	7.0	187
17	37	7.0	188
18	48	5.4	203
19	44	6.0	198
20	47	5.4	202

[Backup] Raw Data

100 V			
#	Flow Rate (LPM)	Pressure (psi)	Power (Watts)
1	70	1	235
2	0	10.5	146
3	20	9	167.4
4	30	8	175.1
5	35	7.5	178.8
6	40	7	184.4
7	45	6	190.5
8	50	5	198.1
9	55	4.25	208
10	48	5.25	195.2
11	44	6.25	188.8
12	40	7	183.3
13	37	7.25	179.6
14	22	8.75	171.2
15	37	7.75	176
16	37.5	6.5	187.4
17	50	5.75	196.6
18	54	4.25	201
19	49	5.5	193.6
20	39	7.25	181.5

125 V			
#	Flow Rate (LPM)	Pressure (psi)	Power (Watts)
1	76	1.5	305
2	0	10.5	254
3	27	8.8	265
4	35	8	268
5	45	6.6	275
6	47	6	277
7	52	5.4	280
8	55	5	282
9	57	4.8	284
10	20	9.0	263
11	38	7.6	270
12	43	7	272
13	20	9.2	263
14	21	8.8	266
15	30	8.2	268
16	40	7.4	270
17	45	6.5	275
18	50	5.7	278
19	43	6.8	272
20	52	4.7	282

[Backup] Sample Calculation of Head

Using one 100V datapoint

Flow Q = 40 LPM

Gauge Pressure P = 7 psi

Power $P_{elec} = 7psi$

Centerline height $z_{cl} = 10.25in = 0.26m$

Converting pressure psi to Pa

$$\Delta p = 7\ psi \times \frac{6894.76\ Pa}{1\ psi} = 48263.32\ Pa$$

Computing dynamic head

$$H_p = \frac{\Delta p}{\rho g} + z_{cl} = \frac{48263.32\ Pa}{\left(\frac{1000kg}{m^3}\right)\left(\frac{9.81m}{s^2}\right)} + 0.26m = 5.18m$$

[Backup] Sample Calculation of Efficiency

Converting flow LPM to $\frac{m^3}{s}$

$$\mathrm{Q} = 40\mathrm{LPM} \times \frac{1\mathrm{m}}{60\mathrm{s}} \times \frac{1m^3}{1000\ L} = \frac{0.0006667m^3}{s}$$

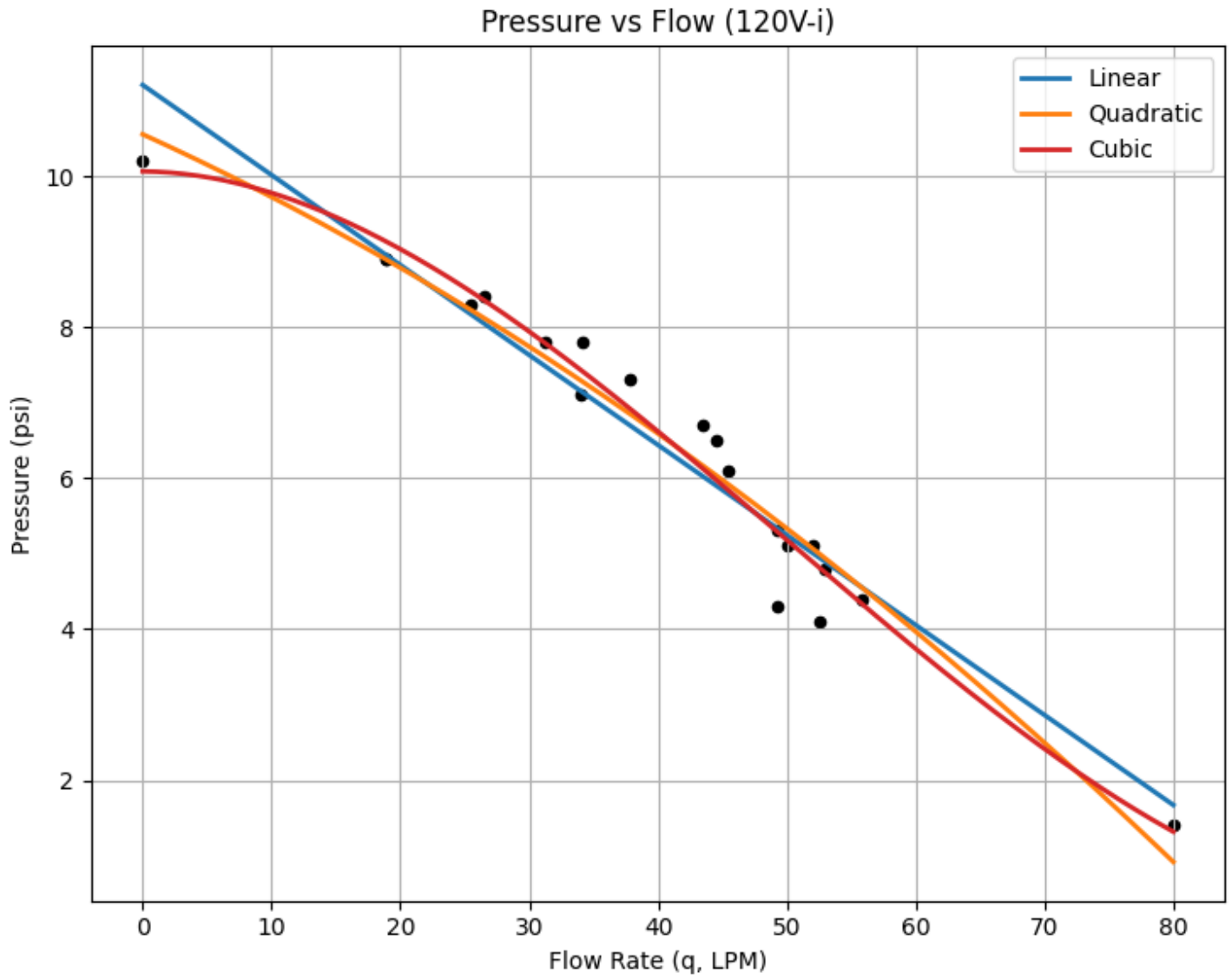
Computing hydraulic power

$$P_{hyd} = \rho g Q H_p = \left(\frac{1000kg}{m^3}\right)\left(\frac{9.81m}{s^2}\right)\left(\frac{0.0006667m^3}{s}\right)(5.18m) = 33.9W$$

Computing Efficiency

$$\eta = \frac{P_{hyd}}{P_{elec}} = \frac{33.9W}{203W} = 16.7\%$$

[Backup] Pressure vs Flow Regression Types

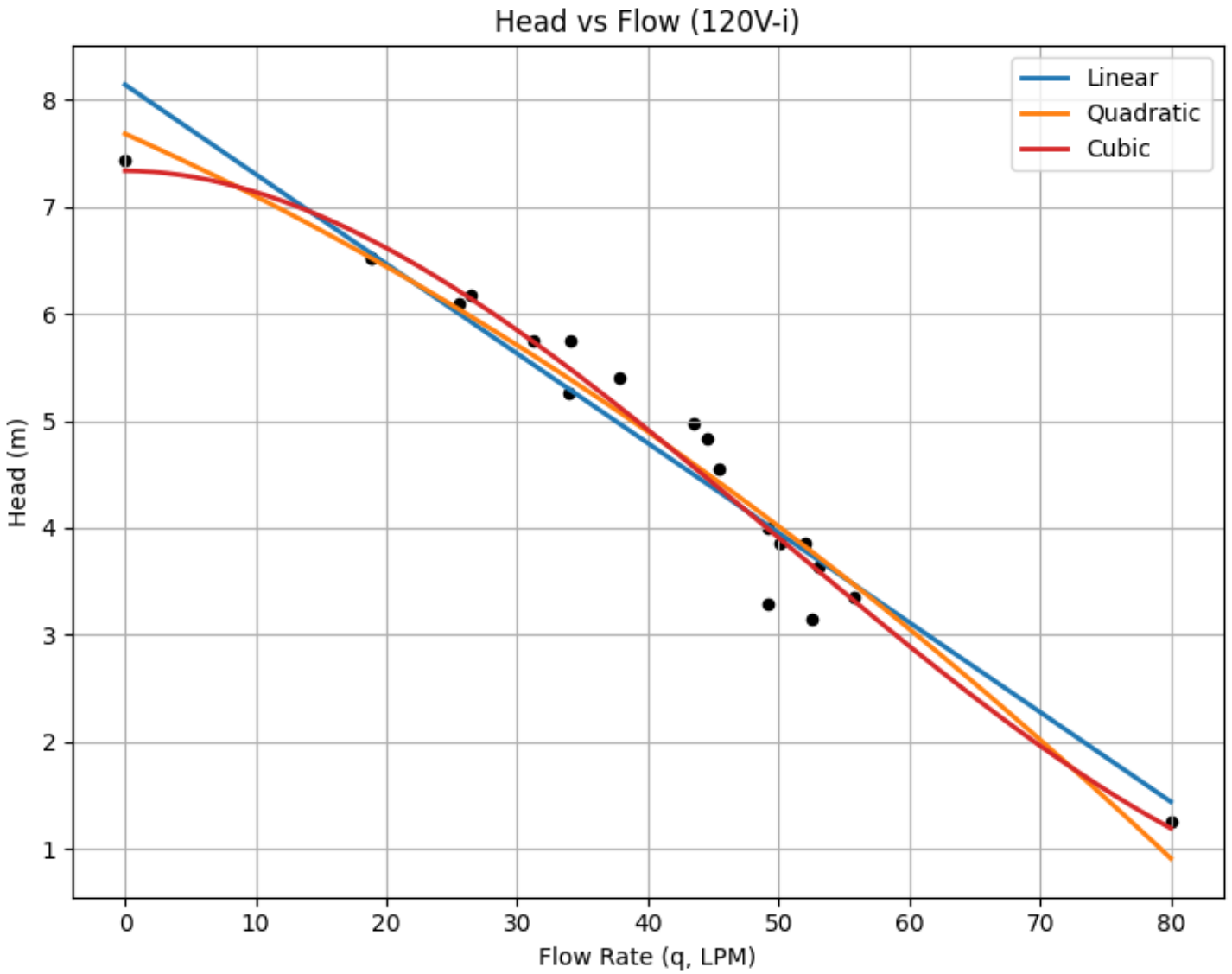


Linear:
$R^2 = 0.9430$
$-0.1192q + 11.21$

Quadratic:
$R^2 = 0.9567$
$-0.000535q^2 + -0.07768q + 10.55$

Cubic:
$R^2 = 0.9676$
$1.918e\text{-}05q^3 + -0.002882q^2 + -0.001535q + 10.06$

[Backup] TDH vs Flow Regression Types

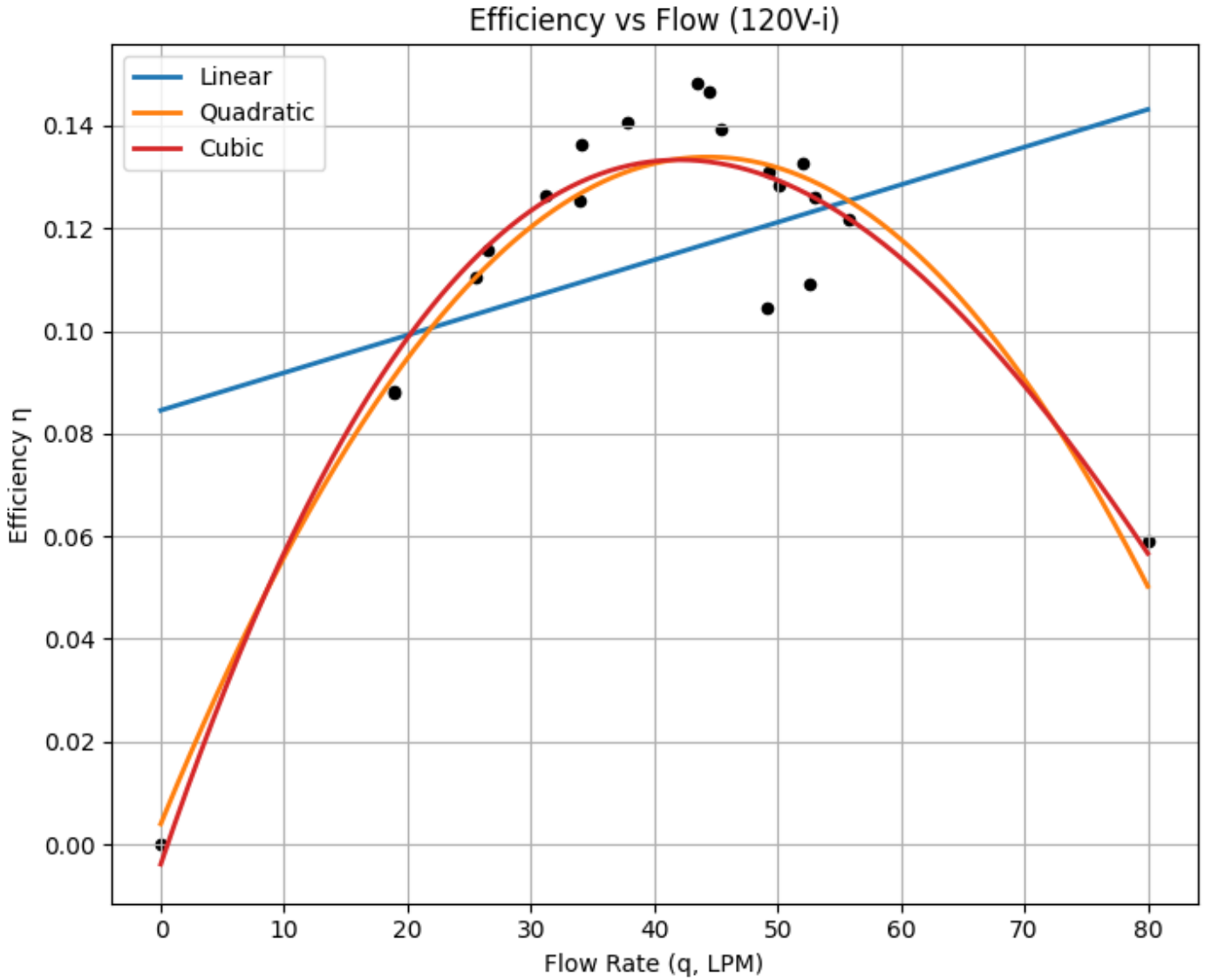


Linear:
$R^2 = 0.9430$
$-0.0838q + 8.144$

Quadratic:
$R^2 = 0.9567$
$-0.000376q^2 + -0.05459q + 7.685$

Cubic:
$R^2 = 0.9676$
$1.348e\text{-}05q^3 + -0.002025q^2 + -0.001079q + 7.34$

[Backup] Efficiency vs Flow Regression Types

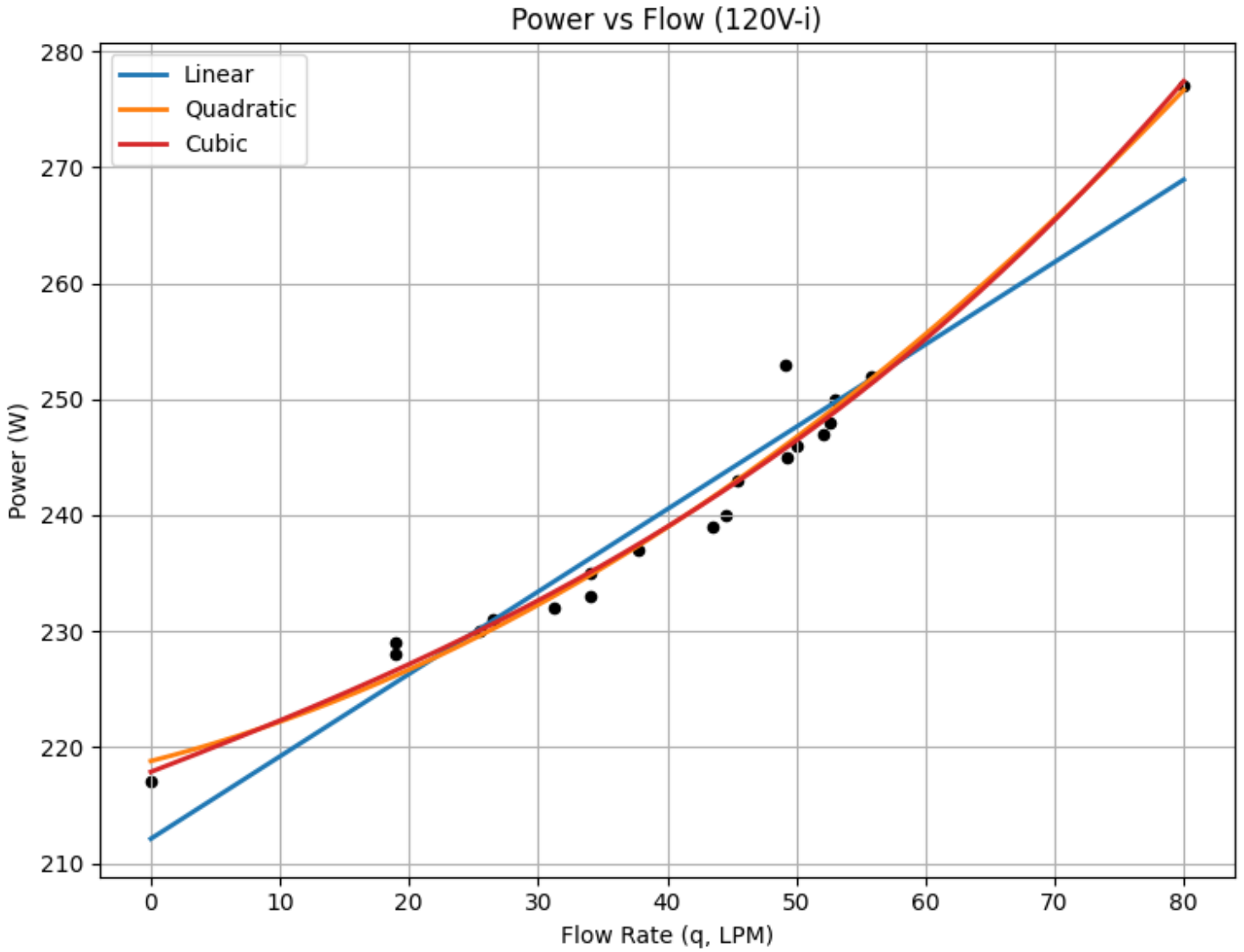


Linear:
$R^2 = 0.1334$
$0.0007325q + 0.08453$

Quadratic:
$R^2 = 0.9152$
$-6.594e\text{-}05q^2 + 0.005854q + 0.003964$

Cubic:
$R^2 = 0.9256$
$3.066e\text{-}07q^3 + -0.0001035q^2 + 0.007071q + -0.003887$

[Backup] Power vs Flow Regression Types



Linear:
$R^2 = 0.9320$
$0.7103q + 212.1$

Quadratic:
$R^2 = 0.9721$
$0.005486q^2 + 0.2842q + 218.8$

Cubic:
$R^2 = 0.9733$
$3.66e\text{-}05q^3 + 0.001007q^2 + 0.4295q + 217.9$